This accessibility comes with trade-offs. Mesh models approximate complex shapes through networks of points and connecting lines, inherently sacrificing precision for performance. Increasing mesh density improves accuracy but demands greater computational resources, creating a constant balance between visual fidelity and system responsiveness. Despite these limitations, mesh-based models excel in their primary role of intuitive visualization. They prove especially valuable in browser-based digital twin applications, where efficient rendering of realistic environments enables users to quickly understand and navigate complex physical assets without requiring specialized software or high-end hardware.

PARAMETRIC 3D MODELS

Parametric 3D models define geometry through mathematical relationships, parameters, and constraints rather than explicit coordinate data like mesh based models. A cylinder in a parametric model may be defined by its radius and height, and the relationship between these, for example, that the height of the cylinder is three times the radius. The dimensions of the geometric object may change, but this relationship will be maintained in a parametric model. This type of model can easily be iterated on by simply modifying parameters without having to rebuild the whole model, while maintaining the intent of the design during modifications. Parametric modelling is popular and essential in many industries where the accuracy and precision of complex designs are essential, like aerospace, automotive, and mechanical engineering. Most modern professional *computer aided design (CAD)* software is primarily parametric.

Parametric 3D models provide capabilities within digital twins that extend beyond those offered by mesh based geometric modelling. If you are building a design driven digital twin where it is important to model the underlying engineering design that governs a structure or system, or you need to precisely model constraints such as clearances, alignment, and dimensions, then parametric models are essential. Parametric models also have the ability to be modified dynamically based on sensor readings or reconfigured to execute different simulations. Furthermore, parametric models can represent entire product families, automatically generating variants for different applications. A CAD model of an offshore oil platform